

Installing and Using WSL2 on Windows

Setting up **WSL2** (Windows Subsystem for Linux 2) and **Windows Terminal**

What You'll Install

1. **WSL2** - Run Linux directly on Windows
2. **Windows Terminal** - Modern terminal application
3. **Ubuntu** - The Linux distribution we'll use

What is WSL2?

- **Windows Subsystem for Linux version 2**
- Lets you run a real Linux environment on Windows
- No need for dual boot or virtual machine
- Full Linux kernel with better performance
- Works great with development tools

WSL2 is not Linux Emulation, it's a real Linux kernel running on Windows!

- Runs real Linux binaries on Real Linux kernel
- Full system call compatibility
- Supports Docker and other advanced tools
- Seamless file sharing between Windows and Linux
- It's just like having Linux installed on your machine!

For ASE courses and projects, WSL2 is the recommended way to run Linux on Windows!

- UNIX/Linux is widely used in software development
- Use Linux command line tools
- Run development environments natively
- Run scripts and programs for automation
- All examples and instructions will assume Linux or Mac environment (not Windows CMD or PowerShell)

Prerequisites

Before starting, make sure you have:

-  Windows 10 (Version 2004 or higher) **OR** Windows 11
-  Administrator access on your computer
-  Internet connection
-  At least 4GB of free disk space

Step 1: Install WSL2

Quick Installation (Recommended)

1. Open PowerShell as Administrator

- Press Windows Key
- Type "PowerShell"
- Right-click → "Run as administrator"

2. Run this command:

```
wsl --install
```

Note: This command automatically installs WSL2 (not WSL1) on Windows 10 version 2004+ and Windows 11.

3. Wait for installation to complete (may take 5-10 minutes)

Step 2: Restart Your Computer

- After installation finishes, **restart your computer**
- This is required for WSL2 to work properly

Step 3: Set Up Ubuntu

After restarting:

1. Ubuntu will launch automatically

- If not, search for "Ubuntu" in Start Menu

2. Create your Linux username and password

- Enter a username (lowercase, no spaces)
- Enter a password (you won't see it as you type - this is normal!)
- Re-enter password to confirm

 **Important:** Remember this password! You'll need it for `sudo` commands.

Step 4: Verify WSL2 Installation

1. Open PowerShell (doesn't need to be as Administrator)
2. Check your installation:

```
wsl -l -v
```

3. Expected output:

NAME	STATE	VERSION
* Ubuntu	Running	2

✓ Make sure VERSION shows **2** (not 1)

If VERSION Shows 1

If your Ubuntu shows VERSION 1, upgrade it:

```
wsl --set-version Ubuntu 2
```

Set WSL2 as default for future installations:

```
wsl --set-default-version 2
```

Install Windows Terminal

Method 1: Microsoft Store (Easiest)

1. Open Microsoft Store
2. Search for "Windows Terminal"
3. Click **Install**

Method 2: Using PowerShell

```
winget install --id Microsoft.WindowsTerminal -e
```

Why Windows Terminal?

- **Multiple Tabs** - Run Linux, PowerShell, and CMD in one window
- **Split Panes** - Work in multiple terminals side-by-side
- **Customizable** - Change colors, fonts, and themes
- **Better Performance** - GPU-accelerated text rendering
- **Modern Interface** - Clean and easy to use

Using WSL2 in Windows Terminal

Launch Ubuntu in Windows Terminal

1. Open Windows Terminal
2. Click the ▼ (down arrow) next to the + tab
3. Select "Ubuntu" from the dropdown menu

Now you're running Linux inside Windows Terminal!

Basic WSL2 Commands

Check WSL version:

```
wsl -l -v
```

Start WSL:

```
wsl
```

Stop all WSL instances:

```
wsl --shutdown
```

Update WSL:

```
wsl --update
```


Accessing Files Between Windows and Linux

WSL2 is an isolated environment, so all the file systems in WSL2 are separate from Windows by default.

However, Microsoft provides **two ways** to access files across the boundary:

1. **From Linux → Access Windows files** via `/mnt/`
2. **From Windows → Access Linux files** via `\\wsl$\\` or `\\wsl.localhost\\`

Understanding File Systems

Two Separate File Systems:

- **Windows File System:** `C:\` , `D:\` , etc.
 - Your Windows files live here
 - Slower when accessed from Linux
- **Linux File System:** `/home/` , `/var/` , etc.
 - Your Linux files live here
 - Faster when working in Linux
 - Accessed from Windows via special network path

From Linux → Access Windows Files

Windows drives are **automatically mounted** at `/mnt/` :

```
# Go to C: drive
cd /mnt/c/

# Go to your Windows user folder
cd /mnt/c/Users/YourUsername/

# List Windows Desktop files
ls /mnt/c/Users/YourUsername/Desktop/

# Access D: drive (if you have one)
cd /mnt/d/
```

Example: Working with Windows Files from Linux

```
# Create a file on Windows Desktop from Linux
echo "Hello from Linux!" > /mnt/c/Users/YourUsername/Desktop/test.txt

# Copy a Windows file to Linux home directory
cp /mnt/c/Users/YourUsername/Documents/file.txt ~/

# Navigate to your Windows Downloads folder
cd /mnt/c/Users/YourUsername/Downloads/
```

 **Note:** Windows & Linux WSL are separated file systems and are connected via a virtual network. Accessing Windows files `/mnt/c/` from Linux is slower than working in Linux filesystem!

From Windows → Access Linux Files

Method 1: File Explorer (Easiest)

1. Open **File Explorer**
2. Look for "**Linux**" (penguin icon) in the left sidebar
3. Click to browse your Ubuntu files
4. Navigate to `Ubuntu` → `home` → `yourusername`

From Windows → Access Linux Files

Method 2: Direct Path in File Explorer

Type in the address bar:

```
\\wsl.localhost\Ubuntu\home\yourusername\
```

Or the shorter version:

```
\\wsl$\\Ubuntu\home\yourusername\
```

 **Note:** wsl\$ is a special network share that gives access to all your WSL distributions!

 **Note:** The `Ubuntu` part may differ if you installed a different distribution.

Example: Opening Linux Files in Windows Apps

From **inside Ubuntu terminal**, you can:

```
# Open current directory in Windows File Explorer  
explorer.exe .
```

```
# Open a specific file in default Windows app  
explorer.exe myfile.txt
```

```
# Open VS Code in current directory  
code .
```

The `.exe` extension tells WSL to run a Windows program!

Where Should You Store Your Files?

 **RECOMMENDED: Store projects in Linux filesystem**

```
# Store your code here (FAST)
~/projects/my-app/
# Full path: /home/yourusername/projects/my-app/
```

Why? Much faster file operations when working in Linux!

Where Should You Store Your Files?

✗ NOT RECOMMENDED: Storing projects in Windows filesystem

```
# Storing here is SLOW when using Linux tools  
/mnt/c/Users/YourUsername/projects/my-app/
```

Why? Significant performance penalty when Linux tools access Windows files.

Exception: You can use `/mnt/c/` for sharing files between Windows and Linux occasionally.

File Access Best Practices

Rule: Do not edit Linux files directly from Windows apps!

Why? It can cause file corruption and data loss! Don't forget that accessing Linux files from Windows is done over a virtual network share, which may not handle Linux file attributes properly.

✓ DO:

- Keep your projects in Linux filesystem (`~/projects/`)
- Use `git clone` inside WSL2 (not in `/mnt/c/`)
- Install development tools inside Ubuntu
- Use `/mnt/c/` only to access existing Windows files
- Use `explorer.exe .` to open Linux folders in Windows

✗ DON'T:

- **Don't work on code in `/mnt/c/`** when possible (performance)
- **Don't edit Linux files directly from Windows apps** (can cause corruption)
- **Don't create symlinks** from Windows to Linux (may not work)
- **Don't forget** that file permissions work differently
- **Don't access Git repos** stored in Windows from Linux or vice versa

File Name Translations

Inside Ubuntu terminal, accessing Windows files involves some name translations:

- c: → `/mnt/c/`
- d: → `/mnt/d/`

Inside Windows File Explorer, accessing Linux files involves:

- Linux root `/` → `\\wsl$\\Ubuntu\\`
- Linux home `/home/username/` → `\\wsl$\\Ubuntu\\home\\username\\`

Understanding File Permissions

Linux and Windows handle file permissions differently:

In Linux (WSL2):

```
# Check file permissions
ls -la myfile.txt

# Change permissions
chmod 755 myscript.sh

# Change ownership
chown username:username myfile.txt
```

Note: Files in `/mnt/c/` may have different permission behavior!

Common File Access Scenarios

Scenario 1: Clone a GitHub repo

```
#  CORRECT: Clone to Linux filesystem  
cd ~  
git clone https://github.com/username/repo.git
```

```
#  WRONG: Don't clone to Windows filesystem  
cd /mnt/c/Users/YourUsername/  
git clone https://github.com/username/repo.git # SLOW!
```

Common File Access Scenarios

Scenario 2: Share a file between Windows and Linux

```
# Copy from Windows to Linux
```

```
cp /mnt/c/Users/YourUsername/Desktop/data.csv ~/
```

```
# Copy from Linux to Windows
```

```
cp ~/results.txt /mnt/c/Users/YourUsername/Desktop/
```


Common File Access Scenarios

Scenario 3: Edit Linux files with Windows VS Code

```
# From inside Ubuntu terminal  
cd ~/projects/my-app  
code .
```

✅ VS Code will open and edit files on WSL2 safely using the **WSL Remote extension**.

⚠️ Not Recommended:

Opening WSL files in Windows apps for editing.

✓ Reading is OK

✗ Editing & saving can break permissions, Git, and file watchers.

(Almost) Perfect Linux Environment on Windows using GitHub/VSCode/Windows Terminal/Explorer.exe and WSL2

With these tools, we can use **WSL2 Ubuntu** as the main development environment!

1. Use **Windows Terminal** for a great terminal experience.
2. Use **VS Code WSL Remote extension** to edit code directly in WSL2 & Managing files.
3. Use **Explorer.exe** to open Linux folders in Windows Explorer when needed.
4. Use **GitHub** for version control and collaboration.

Quick File Access Commands

Show current directory path

`pwd`

List files with details

`ls -la`

Go to home directory

`cd ~`

or

`cd /home/yourusername/`

Go to Windows Desktop from Linux

`cd /mnt/c/Users/YourUsername/Desktop/`

Open current Linux directory in Windows Explorer
`explorer.exe .`

Common Issues and Solutions

Issue: "WSL 2 requires an update to its kernel component"

Solution:

1. Download the WSL2 kernel update from:
<https://aka.ms/wsl2kernel>
2. Run the installer
3. Restart your computer

Common Issues and Solutions

Issue: "The attempted operation is not supported"

Solution:

- Make sure Virtualization is enabled in BIOS
- Check Windows version: Run `winver` and ensure version 2004+

Issue: Ubuntu is slow

Solution:

```
wsl --shutdown
```

Then restart Ubuntu

Issue: Can't find files I created in Windows

Solution:

Check the correct `/mnt/` path:

```
# Windows C:\Users\YourName\Desktop\file.txt is at:  
ls /mnt/c/Users/YourName/Desktop/file.txt
```

Note: Replace `YourName` with your actual Windows username!

Issue: Permission denied when editing files

Solution:

```
# If file is owned by root, change ownership  
sudo chown $USER:$USER myfile.txt  
  
# Or run with sudo (be careful!)  
sudo nano myfile.txt
```

Updating Ubuntu

Inside Ubuntu terminal, run:

```
# Update package list  
sudo apt update  
  
# Upgrade installed packages  
sudo apt upgrade -y
```

Run this periodically to keep your system updated.

Installing Common Development Tools

Use `apt` to install tools like Git, Node.js, Python, etc.

- The `-y` flag auto-confirms prompts

```
apt install git nodejs python3 -y
```

Enjoy UNIX/Linux Environment on Windows!

Most developers prefer UNIX/Linux for development because:

- You can use any shell such as zsh or bash
- You can make scripts and automate tasks easily using the shell scripts
- You can make python scripts and run them natively
- You can use Docker and other container tools natively

You can do all these things on WSL2!